

Q1 (25 July 2021 Shift 1)

A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

$$\varepsilon(x) = \varepsilon_0 + kx, \text{ for } \left(0 < x \leq \frac{d}{2}\right)$$

$$\varepsilon(x) = \varepsilon_0 + k(d - x), \text{ for } \left(\frac{d}{2} \leq x \leq d\right)$$

(1) $\left(\varepsilon_0 + \frac{kd}{2}\right)^{2/kA}$

(2) $\frac{kA}{2 \ln\left(\frac{2\varepsilon_0 + kd}{2\varepsilon_0}\right)}$

(3) 0

(4) $\frac{kA}{2} \ln\left(\frac{2\varepsilon_0}{2\varepsilon_0 - kd}\right)$

Q2 (25 July 2021 Shift 2)

If q_f is the free charge on the capacitor plates and q_b is the bound charge on the dielectric slab of dielectric constant k placed between the capacitor plates, then bound charge q_b can be expressed as :

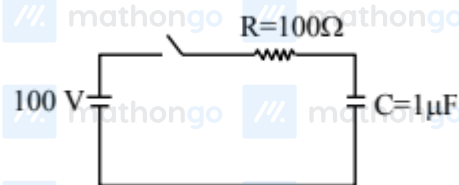
(1) $q_b = q_f \left(1 - \frac{1}{\sqrt{k}}\right)$

(2) $q_b = q_f \left(1 - \frac{1}{k}\right)$

(3) $q_b = q_f \left(1 + \frac{1}{\sqrt{k}}\right)$

(4) $q_b = q_f \left(1 + \frac{1}{k}\right)$

Q3 (27 July 2021 Shift 1)



A capacitor of capacitance $C = 1\mu\text{F}$ is suddenly connected to a battery of 100 volt through a resistance

$R = 100\Omega$. The time taken for the capacitor to be charged to get 50 V is :

[Take $\ln 2 = 0.69$]

Questions with Answer Keys

MathonGo

(1) $1.44 \times 10^{-4} \text{ s}$

(2) $3.33 \times 10^{-4} \text{ s}$

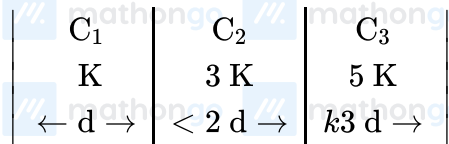
(3) $0.69 \times 10^{-4} \text{ s}$

(4) $0.30 \times 10^{-4} \text{ s}$

Q4 (27 July 2021 Shift 1)

In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be :

(Given area of plate = A)



(1) $\frac{15}{34} \frac{K\epsilon_0 A}{d}$

(2) $\frac{15}{6} \frac{K\epsilon_0 A}{d}$

(3) $\frac{25}{6} \frac{K\epsilon_0 A}{d}$

(4) $\frac{9}{6} \frac{K\epsilon_0 A}{d}$

Q5 (27 July 2021 Shift 1)

Two capacitors of capacities $2C$ and C are joined in parallel and charged up to potential V . The battery is removed and the capacitor of capacity C is filled completely with a medium of dielectric constant K . The potential difference across the capacitors will now be :

(1) $\frac{V}{K+2}$

(2) $\frac{V}{K}$

(3) $\frac{3V}{K+2}$

(4) $\frac{3V}{K}$

Answer Key

Q1 (2)

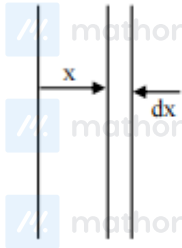
Q2 (2)

Q3 (3)

Q4 (1)

Q5 (3)

Q1



Taking an element of width dx at a distance x ($x < d/2$) from left plate

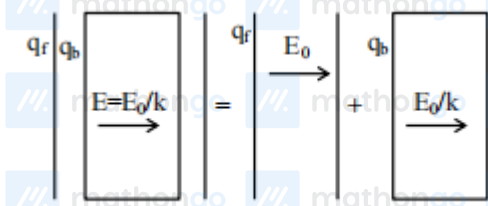
$$dc = \frac{(\epsilon_0 + kx)A}{dx}$$

$$\text{Capacitance of half of the capacitor } \frac{1}{C} = \int_0^{d/2} \frac{1}{dc} = \frac{1}{A} \int_0^{d/2} \frac{dx}{\epsilon_0 + kx}$$

$$\frac{1}{C} = \frac{1}{kA} \ln\left(\frac{\epsilon_0 + kd/2}{\epsilon_0}\right)$$

$$\text{Capacitance of second half will be same } C_{eq} = \frac{C}{2} = \frac{kA}{2 \ln\left(\frac{2\epsilon_0 + kd}{2\epsilon_0}\right)}$$

Q2



When a dielectric is inserted in a capacitor Due to free charge $\vec{E} = \vec{E}_0$ only After dielectric $E' = \frac{E_0}{k}$

$$q_B = q_f \left(1 - \frac{1}{k}\right)$$

Q3

$$V = V_0 \left(1 - e^{-\frac{t}{RC}}\right)$$

$$50 = 100 \left(1 - e^{-\frac{t}{RC}}\right)$$

$$t = 0.69 \times 10^{-4} \text{ sec.}$$

Q4

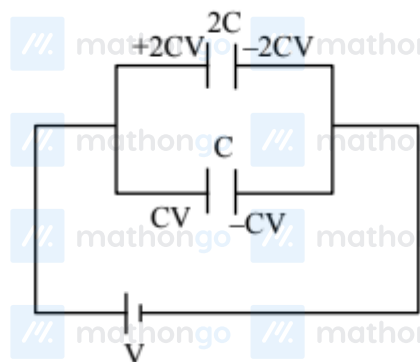
Hints and Solutions

MathonGo

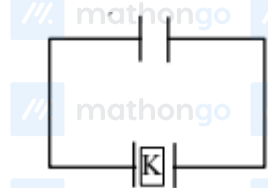
$$\frac{1}{C_{\text{eff}}} = \frac{d}{K\epsilon_0 A} + \frac{2d}{3K\epsilon_0 A} + \frac{3d}{5K\epsilon_0 A}$$

$$C_{\text{eff}} = \frac{15K\epsilon_0 A}{34d}$$

Q5



Now,



$$V_C = \frac{2CV + CV}{KC + 2C}$$

$$= \frac{3V}{K + 2}$$